

AUTOM-99: Best Practice Summary

Series: AUTOM — Dynatrace Automation | **Notebook:** 99 | **Created:** March 2026 |
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This notebook consolidates every actionable best practice from the AUTOM series (notebooks 01-09) into a single reference. Each practice is definitive: it tells you exactly what to set, not what to consider.

Use this as a checklist when designing, implementing, or auditing Dynatrace configuration automation.

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Prerequisites

Requirement	Details
AUTOM Series	Familiarity with AUTOM-01 through AUTOM-09
Dynatrace Environment	SaaS tenant with admin access
Automation Tooling	Monaco CLI, Terraform CLI, or both installed

1. Authentication & Token Management

Practice	Recommended Setting/Value	Priority
Use least-privilege scopes	Grant only <code>settings.read</code> , <code>settings.write</code> , <code>ReadConfig</code> , <code>WriteConfig</code> for config tools. Add <code>ExternalsSyntheticIntegration</code> only when managing synthetics.	Critical
Configure a platform credential and a classic API token	A Platform Token or an OAuth client covers Settings 2.0 and platform resources (Workflows, Documents, Segments); with the Terraform provider set <code>DYNATRACE_HTTP_OAUTH_PREFERENCE=true</code> when using a Platform Token for those. A classic API Token (<code>dt0c01.*</code>) is still required for Synthetics and <code>dynatrace_slo_v2</code> . Account IAM resources need an OAuth client (<code>client_id</code> , <code>client_secret</code> , <code>account_id</code>). In Monaco, reference them as <code>auth.token</code> plus <code>auth.platformToken</code> or <code>auth.oAuth</code> — access and platform tokens are not interchangeable.	Critical
Never hardcode tokens	Store tokens in environment variables, HashiCorp Vault, AWS Secrets Manager, or equivalent secret store.	Critical
Separate tokens per environment	Create distinct tokens for dev, staging, and production tenants. Never share tokens across environments.	Critical

Practice	Recommended Setting/Value	Priority
Rotate tokens on a schedule	Set a rotation cadence (e.g., 90 days). For short-lived tokens, retrieve from Vault at pipeline runtime.	Recommended
Prefer Platform Tokens over classic Access Tokens	Platform Tokens scope to the user's existing permissions and simplify token management.	Recommended
Authenticate pipelines as a service user	Issue a platform token for a service user (AUTOM-96), or create an OAuth client whose subject user is that service user — never a person's own credentials. Account IAM resources need an OAuth client: platform tokens can't be used for IAM (Account Management).	Recommended

Sources: [OAuth clients \(DT docs\)](#) — "This needs to be an active user, and either a service user or any user with account-user-management permission."; [Platform tokens \(DT docs\)](#) — "They can be assigned to the user creating them or to a service user that the creating user has access to."; [Provider configuration \(Dynatrace GitHub\)](#) — "Platform tokens can't be used for IAM (Account Management) or classic resources."

2. Settings API

Practice	Recommended Setting/Value	Priority
Use Settings 2.0 API for all new configuration	Endpoint: <code>/api/v2/settings/objects</code> . Do not use Classic Config API for new configs.	Critical

Practice	Recommended Setting/Value	Priority
Discover schemas before creating objects	Call <code>GET /api/v2/settings/schemas</code> to find the correct <code>schemaId</code> , then <code>GET /api/v2/settings/schemas/{schemaId}</code> for field requirements.	Critical
Use bulk POST for multiple objects	Send an array of objects in a single <code>POST /api/v2/settings/objects</code> call instead of individual requests.	Recommended
Check before create (idempotency)	Query for existing objects by <code>schemaIds</code> and <code>scopes</code> before creating. Use <code>PUT</code> for updates (idempotent), not <code>POST</code> .	Recommended
Implement exponential backoff on 429	Retry with increasing delay when rate-limited, and throttle bulk operations client-side. Dynatrace documents no fixed request rate: 429 means the environment's request thread pool and its queue are full, so the limit depends on load.	Recommended
Track object IDs for lifecycle management	Store returned <code>objectId</code> values for subsequent update and delete operations.	Recommended
Review existing objects as examples	Before writing new config, <code>GET /api/v2/settings/objects?schemaIds=<schema></code> to see the structure of existing objects in your tenant.	Optional

Sources: [Access limit \(DT docs\)](#) — "You reach the limit when both thread pool and its queue are full or the request times out in the queue (timeout is 10 seconds)."

3. Monaco Configuration-as-Code

Practice	Recommended Setting/Value	Priority
Always run <code>monaco deploy --dry-run</code> before deploy	Monaco has no <code>validate</code> command. <code>monaco deploy manifest.yaml --dry-run</code> parses YAML, checks template JSON and resolves references without contacting the tenant — it cannot catch payload errors, so a deploy can still fail with HTTP 400 after a clean dry-run.	Critical
Use environment variables for auth	Set <code>DT_TENANT_URL</code> and <code>DT_API_TOKEN</code> as env vars. In <code>manifest.yaml</code> , <code>auth.token.name</code> holds the variable name , never the token.	Critical
Use meaningful config IDs	IDs should describe the config purpose (e.g., <code>production-segment</code> , <code>web-app-problem-routing</code>). These IDs are how Monaco tracks objects.	Critical
Use <code>manifest.yaml</code> for multi-environment setup	Define all environments in <code>environmentGroups</code> with separate URL/token env vars per environment.	Recommended
Use environment-specific overrides	Add <code>environmentOverrides</code> (or <code>groupOverrides</code>) at the config level, sibling of <code>config:</code> , to vary values (e.g., <code>delay_minutes: 1</code> for prod, <code>5</code> for staging).	Recommended
Use <code>skip</code> with overrides for env-specific configs	<code>skip</code> is a boolean on <code>config:</code> ; set a default and flip it per environment or environment group in <code>environmentOverrides</code> / <code>groupOverrides</code> .	Recommended
Use projects to group related configs	Put related configs in one project and deploy it with <code>--project</code> . <code>--group</code> selects manifest environment groups , not configs; there is no per-config <code>group:</code> key.	Recommended
Use reference parameters for ordering	Reference other configs with <code>["<project>", "<configType>", "<configId>", "id"]</code> (for settings, <code>configType</code> is the schema ID) so Monaco deploys the dependency first.	Recommended

Practice	Recommended Setting/Value	Priority
Modular directory structure	Separate configs by domain: <code>segments/</code> , <code>workflows/</code> , <code>slo-v2/</code> (classic estates: <code>management-zones/</code> , <code>alerting-profiles/</code> until the upgrade).	Recommended
Download before creating from scratch	Run <code>monaco download</code> to get the current state, then modify the exported YAML.	Optional

4. Terraform Infrastructure-as-Code

Practice	Recommended Setting/Value	Priority
Pin the provider version	<code>version = "~> 1.105"</code> in <code>required_providers</code> block (AUTOM-04), and commit <code>.terraform.lock.hcl</code> so every run uses the same provider build (AUTOM-96). Never use unpinned versions.	Critical
Use remote state storage	Configure <code>backend "s3"</code> or <code>terraform.cloud</code> block. Never use local state for teams.	Critical
Enable state locking	Use the S3 backend with <code>use_lockfile = true</code> (Terraform 1.11+; DynamoDB-based locking is deprecated), HCP Terraform, or an equivalent locking backend to prevent concurrent applies.	Critical
Always run <code>terraform plan</code> before <code>terraform apply</code>	Review the plan output for every change. Automate plan-as-PR-comment in CI.	Critical
Use dual auth for full resource coverage	Configure both <code>dt_api_token</code> (for Synthetics and the classic <code>dynatrace_slo_v2</code>) and a platform token or OAuth client (for Gen3 resources, including <code>dynatrace_platform_slo</code> ; OAuth <code>client_id</code> / <code>client_secret</code> / <code>account_id</code> for IAM). The provider routes each resource to the correct auth method.	Critical

Practice	Recommended Setting/Value	Priority
Use modules for reusable patterns	Create modules for standard patterns (e.g., <code>modules/slo/</code> , <code>modules/problem-routing/</code> — see AUTOM-09 §5; classic <code>modules/alerting-profile/</code> - style modules only while the tenant is unupgraded).	Recommended
Use variable validation in modules	Add <code>validation</code> blocks to enforce naming, environment values, and delay constraints at module input.	Recommended
Use a directory per environment	<code>envs/dev/</code> , <code>envs/staging/</code> , <code>envs/prod/</code> , each with its own backend and credentials (AUTOM-09 §6). CLI workspaces are not appropriate when environments need separate credentials and access controls.	Recommended
Import existing resources into state	Write the HCL block first, then <code>terraform import <resource> "<object-id>"</code> . Never recreate what already exists.	Recommended
Use <code>terraform state rm</code> to detach without deleting	When you need to stop managing a resource without destroying it.	Recommended
Format code with <code>terraform fmt</code>	Run before every commit for consistent HCL formatting.	Recommended
Use the export utility for brownfield adoption	Run <code>DYNATRACE_TARGET_FOLDER=./exported ./terraform-provider-dynatrace -export -ref -id</code> to generate HCL from existing configs (the output folder comes from <code>DYNATRACE_TARGET_FOLDER</code> ; there is no <code>-target-folder</code> flag).	Optional

Sources: [S3 backend \(Terraform docs\)](#) — "DynamoDB-based locking is deprecated and will be removed in a future minor version."; [Terraform v1.11.0 release notes \(HashiCorp GitHub\)](#) — "S3 native state locking is now generally available."; [Workspaces \(Terraform docs\)](#) — "Workspaces are

not appropriate for system decomposition or deployments requiring separate credentials and access controls."

5. Dynatrace Workflows

Practice	Recommended Setting/Value	Priority
One workflow, one purpose	Each workflow handles a single responsibility (e.g., "problem email notification" not "do everything on problem").	Critical
Make tasks idempotent	Tasks must be safe to run multiple times without side effects (e.g., check if ticket exists before creating).	Critical
Store secrets in Credential Vault	Use Dynatrace Credential Vault for API keys, webhook URLs, and integration tokens. Never hardcode secrets in workflow YAML or JavaScript.	Critical
Scope triggers narrowly	Filter triggers by entity tag (ownership), problem category, severity, and a custom DQL filter — management-zone scoping is classic. Do not trigger on all problems.	Critical
Add retries on external calls	Set the task retry <code>count</code> (1–99) and <code>delay</code> in seconds (1–3600) on HTTP tasks and external integrations — e.g., count 3, delay 30.	Recommended
Route errors to a handler task	Add a handler task whose condition is the upstream task's state (<code>states: {api_call: ERROR}</code> / Terraform <code>conditions { states = { api_call = "ERROR" } }</code>). Workflows have no <code>on_error</code> construct.	Recommended
Validate entity type before remediation	Check <code>entity.type</code> in a JavaScript task before executing remediation (e.g., only restart <code>PROCESS_GROUP_INSTANCE</code> , not hosts).	Recommended
Debounce flapping alerts	Add delays or deduplication checks for alerts that open/close rapidly.	Recommended

Practice	Recommended Setting/Value	Priority
Use manual triggers for testing	Test workflow logic with manual triggers before enabling detected-problem triggers.	Recommended
Log all remediation actions	Ensure every auto-remediation action writes to the Dynatrace audit trail and comments on the originating problem.	Recommended
Use sub-workflows for reusable logic	Extract common patterns (e.g., "create ITSM ticket") into sub-workflows callable from multiple parent workflows.	Optional
Use approval gates for risky remediations	Enable built-in Approval Requests for actions like scaling, restarts, or config changes.	Optional

6. SDKs & Programmatic Access

Practice	Recommended Setting/Value	Priority
Use environment variables for credentials	Set <code>DT_URL</code> and <code>DT_API_TOKEN</code> as environment variables. Never pass tokens as function arguments or config literals.	Critical
Handle paginated responses	Always iterate through all pages. Check for <code>nextPageKey</code> in responses and loop until exhausted.	Critical
Implement rate limiting	Throttle client-side with a rate-limiting decorator/wrapper (AUTOM-06) and back off exponentially on <code>429</code> . Dynatrace documents no fixed request rate — the limit depends on load (§2).	Recommended
Use type-safe SDK clients	Use the TypeScript <code>@dynatrace-sdk/*</code> clients (e.g., <code>@dynatrace-sdk/client-query</code>) in Dynatrace apps and functions. Dynatrace publishes no Python SDK package — Python automation calls the REST APIs directly.	Recommended

Practice	Recommended Setting/Value	Priority
Wrap API calls in error handlers	Catch errors around every call; log the HTTP status and message (<code>requests : raise_for_status()</code>). Re-raise what you cannot handle.	Recommended
Add structured logging	Log request method, endpoint, status code, and duration for every API call.	Recommended
Use MCP Server for AI-assisted access	Connect Dynatrace MCP Server to Claude Code, GitHub Copilot, or Amazon Q for natural-language queries against your tenant.	Optional

7. CI/CD & GitOps

Practice	Recommended Setting/Value	Priority
Store all config in Git	Every Dynatrace configuration (Monaco YAML, Terraform HCL) lives in version control. No exceptions.	Critical
Require PR reviews for production changes	Enable branch protection on <code>main</code> . Require at least 1 approval. Use CODEOWNERS for Dynatrace config paths.	Critical
Mask all tokens in CI/CD	Store tokens as masked secrets in GitHub Actions, GitLab CI, or equivalent. Never echo tokens in logs.	Critical
Validate on every PR	Run <code>monaco deploy --dry-run</code> or <code>terraform validate + terraform plan</code> on every pull request. Post plan output as PR comment.	Critical
Deploy only from main branch	Gate <code>terraform apply</code> and <code>monaco deploy</code> to run only on push to <code>main</code> (not on PR branches).	Critical
Use staged rollout: dev, staging, production	Deploy to dev first, then staging, then production. Require manual approval gate before production.	Critical

Practice	Recommended Setting/Value	Priority
Schedule drift detection	Run <code>terraform plan -detailed-exitcode</code> on a cron schedule (e.g., weekdays 6am UTC). Exit code <code>2</code> means drift. Auto-create GitHub Issue on drift.	Recommended
Use Vault for runtime credentials	Retrieve tokens at pipeline runtime from HashiCorp Vault using OIDC/JWT auth instead of static CI/CD secrets.	Recommended
Use reusable workflows for multi-repo orgs	Define a shared <code>workflow_call</code> workflow that all team repos invoke. Standardizes plan/apply across the organization.	Recommended
Send deployment events to Dynatrace	<code>POST /api/v2/events/ingest</code> with <code>eventType: CUSTOM_DEPLOYMENT</code> , commit SHA, and branch name after every deploy.	Recommended
Use Kustomize overlays for Operator GitOps	Base DynaKube config + per-cluster patches via <code>patches:</code> (<code>patchesStrategicMerge</code> is deprecated in Kustomize). Use <code>apiVersion: dynatrace.com/v1beta6</code> .	Recommended
Encrypt K8s secrets in Git	Use Sealed Secrets, SOPS, or External Secrets Operator. Never commit plaintext Dynatrace tokens to Git.	Critical
Use a PR template for config changes	Include sections: Description, Type of Change, Environments Affected, Validation Checklist, Rollback Plan.	Optional

8. Governance Architecture

Practice	Recommended Setting/Value	Priority
Separate IAM pipeline from config pipeline	Pipeline A (central team) manages <code>dynatrace_iam_*</code> resources. Pipeline B (LOB teams) manages configs under grants from Pipeline A.	Critical

Practice	Recommended Setting/Value	Priority
Single service account writer in production	Only one SA writes to prod. All humans get read-only. Teams develop in dev/test tenant with UI write access.	Critical
Never mix Monaco and Terraform for the same config type	Choose one tool per configuration domain. Mixing causes state conflicts and unpredictable drift.	Critical
Eliminate manual changes alongside automation	If a config is managed by automation, all changes go through the pipeline. No ClickOps in production.	Critical
Always use version control for configs	Every automation artifact (YAML, HCL, JSON) lives in Git with meaningful commit messages.	Critical
Use OPA/Conftest for resource type allowlists	Define a Rego policy that restricts which <code>dynatrace_*</code> resource types each team repo can create.	Recommended
Enforce mandatory team tagging via policy	OPA/Sentinel policy requires every resource to include team ownership metadata (e.g., an <code>owner</code> tag, naming prefix; MZ binding on classic tenants).	Recommended
Lock down Terraform state file access	Restrict HCP Terraform workspace permissions. State files can contain OAuth credentials and IAM binding details.	Recommended
Use brokered self-service for Synthetic monitors	Teams submit declarative requests; a central pipeline owns the environment-wide API token and applies on their behalf. Teams never hold direct API credentials.	Recommended
Use IAM policies for schema-level access	Manage <code>dynatrace_iam_policy</code> resources with <code>statement_query</code> restricting teams to specific <code>settings:objects:*</code> schemas.	Recommended
Document v1 API limitations for auditors	Frame Synthetic v1 API scoping gap as an accepted platform constraint with compensating controls (brokered access, MZ fencing, Sentinel).	Optional
Use module allowlists in Sentinel	Restrict Terraform plans to only approved modules, preventing teams from using unapproved resource patterns.	Optional

9. Migration Automation

Practice	Recommended Setting/Value	Priority
Export before migrating	Run <code>monaco download</code> (with an access token and a platform token, or platform types are skipped with only a log warning) or <code>terraform-provider-dynatrace -export</code> (naming workflows, segments, documents and platform SLOs explicitly — they are excluded by default) on the source tenant. Keep the full backup.	Critical
Re-map hardcoded entity IDs	Entity IDs do not carry over. Use an entity selector only where the target schema has an <code>entitySelector</code> field; otherwise map source IDs to target IDs or parameterise them per environment (AUTOM-08 §3).	Critical
Validate counts after each migration step	Compare per-type config counts between two <code>monaco download</code> s — source and target, same credentials including a platform token (AUTOM-08 §6). Settings API counts cover classic schemas only, and only while both tenants are on the same generation. Note: <code>fetch dt.settings</code> is not valid DQL.	Critical
Test migration in a non-production tenant first	Deploy exported config to a dev/test tenant before touching production.	Critical
Re-enter credentials manually	Credential Vault entries, API tokens, SSL certificates, and cloud integration secrets do not migrate. Re-create them in the target.	Critical
Use Monaco for SaaS-to-SaaS migration	Download from source; write a target manifest with <code>auth.token</code> plus <code>auth.platformToken</code> (or <code>auth.oAuth</code>) whose project path is the downloaded <code>project/</code> folder; <code>monaco deploy --dry-run</code> ; <code>deploy</code> (AUTOM-08 §3).	Recommended

Practice	Recommended Setting/Value	Priority
Use SaaS Upgrade Assistant for Managed-to-SaaS	Install it from Dynatrace Hub in the target SaaS environment, then upload the environment export from the Managed Cluster Management Console (Environments → Export configuration). It tracks progress, flags failed configurations, and supports bulk edit and partial deployment (AUTOM-08 §5).	Recommended
Document non-portable configurations	Keep a checklist of configs that need manual re-creation: credentials, historic data, private locations, cloud integrations.	Recommended
Use Terraform export for brownfield imports	Export existing configs to HCL, then manage via Terraform going forward.	Optional

Sources: [SaaS Upgrade Assistant \(DT docs\)](#) — "Choose which configurations you want to migrate and run partial deployment."; [Manage resources — auth section \(DT docs\)](#) — "Access tokens and platform tokens are not interchangeable."

10. Tool Selection

Use Case	Use This Tool	Priority
One-off configuration change	Settings API (direct REST)	Critical
Repeatable config deployments with GitOps	Monaco	Critical
Full IaC with state management and drift detection	Terraform	Critical

Use Case	Use This Tool	Priority
Event-driven automation and auto-remediation	Dynatrace Workflows	Critical
Custom reporting, complex logic, or integrations	TypeScript <code>@dynatrace-sdk/*</code> (apps, functions) or the REST APIs (Python <code>requests</code>) — there is no official Python SDK	Critical
Tenant-to-tenant migration	Monaco (download/deploy pattern)	Critical
Managed-to-SaaS migration	SaaS Upgrade Assistant	Recommended
AI-assisted querying and triage	Dynatrace MCP Server	Optional
Team already uses Terraform for infrastructure	Terraform (extend existing IaC)	Recommended
Team needs lowest barrier to entry	Monaco (YAML, simple CLI)	Recommended

Anti-Patterns

Anti-Pattern	Consequence	Correct Approach
Mixing Monaco and Terraform for the same config type	State conflicts, unpredictable drift	Choose one tool per config domain
Manual UI changes alongside automation	Configuration drift, lost changes	All changes go through the pipeline
No version control for configs	No rollback, no audit trail	Store everything in Git
Hardcoded secrets in config files	Security breach risk	Use env vars, Vault, or secret managers
Distributing environment-wide write tokens to teams	No access isolation	Use brokered self-service or scoped OAuth

Anti-Pattern	Consequence	Correct Approach
Using <code>fetch dt.metrics</code> in DQL	Invalid command	Use <code>timeseries</code> for metrics

11. Community Resources

Key GitHub repositories organized by tool, providing starter templates, working examples, and hands-on exercises.

Monaco Repositories

Repository	Description
dynatrace-configuration-as-code	Official Monaco CLI (v2.30.0, released 09/23/2026, at time of writing)
community-examples — configuration-as-code	Monaco samples, e.g. <code>basic-templates-monaco</code> , <code>pipeline-observability</code> and <code>access-control</code> configs. Replaces the archived <code>dynatrace-configuration-as-code-samples</code> repo
easytrade	Real-world Monaco project structure (<code>manifest.yaml</code> , detection rules, workflows)
Dynatrace-Config-Manager	Archived, no longer maintained — former GUI tool for tenant-to-tenant config migration; historical reference only
monaco-self-paced-exercises	6 structured hands-on exercises

Terraform Repositories

Repository	Description
terraform-provider-dynatrace	Official provider (v1.105.0 released 09/23/2026 at time of writing — check the registry for newer) with export capability

Repository	Description
community-examples — configuration-as-code	Terraform samples, e.g. <code>basic-templates-terraform</code> , <code>terraform_modules</code> , <code>terraform_dql_example</code> , <code>terraform_team_onboarding</code> , <code>iam_tf_sample</code>

CI/CD & Platform Engineering

Repository	Description
dynatrace-automation-tools	Archived, no longer maintained — former SRG + Events CLI for CI/CD pipelines; its README points to dtctl for similar use cases
platform-engineering-demo	Full IDP reference: ArgoCD + Backstage + Keptn + Dynatrace
demo-crossplane	Crossplane + Terraform GitOps pattern
monaco-demo	Working GitHub Actions workflow for Monaco deploy
obslab-release-validation	Release validation with k6, business events, and SRG

Archive status checked 10/02/2026. `dynatrace-configuration-as-code-samples` , `Dynatrace-Config-Manager` and `dynatrace-automation-tools` are archived. The [samples repo's README \(Dynatrace GitHub\)](#) says "*They now live in Dynatrace Community Examples*", under its `configuration-as-code/` folder — use that folder instead.

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.